

DEEP DIVE · RAG IN PRODUCTION

Your RAG demo works. Production will **break** it.

7 failure modes I design for before a RAG system goes live, and the fix for each.



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swipe →

FAILURE 1 · CHUNKING

The answer gets **split** in half

- Fixed-size chunks cut tables, clauses and steps mid-way
- Chunk by structure: headings, sections, table rows
- Test recall on real user questions, not sample ones



FAILURE 2 · NUMBERS

Vectors doing **math**

- Similarity search finds similar words, not correct totals
- Route metric questions to SQL on the live database
- The database calculates. The model explains.

```
if intent == "metric":  
    rows = run_sql(question)  
    return llm.explain(rows)  
return llm.answer(retrieve(question))
```

FAILURE 3 · FRESHNESS

The index goes **stale**

- Source data changes, the vector index doesn't
- Re-index incrementally with CDC or updated_at watermarks
- Show 'last refreshed' next to every answer



FAILURE 4 · ACCESS

Retrieval ignores **who** is asking

- One shared index means one user can see another's data
- Filter by user and tenant at query time, never in the prompt
- Mirror source permissions, e.g. row-level security



FAILURE 5 · CONTEXT

More chunks, **worse** answers

- Stuffing 20 chunks buries the one that matters
- Retrieve wide, rerank, keep the top 3 to 5
- Fewer tokens: lower cost, lower latency



FAILURE 6 · EVALS

'Looks fine' is not a **test**

- Build a golden set of 50 to 100 real questions
- Score retrieval and answers separately
- Run it on every prompt, model or chunking change



FAILURE 7 · OBSERVABILITY

Failures nobody sees

- Log the query, retrieved chunk ids, answer, latency, cost
- When nothing relevant is found, say so instead of guessing
- Review the lowest-scored answers every week



SUMMARY

Demo RAG answers. Production RAG knows when it can't.

Save this for your next AI build.



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